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Modal Logic in Lean 4

A comprehensive formalization of normal modal propositional logic in Lean 4 with Mathlib, featuring **soundness and completeness proofs for all 16 normal modal logics** in the modal cube, the **finite model property**, and **decidability**.

~11,100 lines of Lean · 0 sorry's · Only standard axioms (propext, Classical.choice, Quot.sound)

Highlights

- **Soundness and completeness** for all **16 distinct** normal modal logics obtainable from subsets of {T, B, 4, D, 5}: K, KT, KB, K4, K5, KD, KTB, S4, KB4, KDB, KD4, KD5, K45, KB5, KD45, S5
- **Finite model property** via filtration for K
- **Decidability** of K-validity (`DecidablePred kValid`)
- **Frame correspondence**: T ↔ Reflexive, B ↔ Symmetric, 4 ↔ Transitive, 5 ↔ Euclidean, D ↔ Serial
- **Undefinability** of irreflexivity
- **Local vs. global consequence** with concrete counterexamples
- **Full axiom audit** via `#print axioms` on every major theorem

The Modal Cube

The project systematically covers every normal modal logic formed by combining five axiom schemas:

Axiom	Schema	Frame Condition
T	$\Box\phi \to \phi$	Reflexivity
B	$\phi \to \Box\Diamond\phi$	Symmetry
4	$\Box\phi \to \Box\Box\phi$	Transitivity
D	$\Box\phi \to \Diamond\phi$	Seriality
5	$\Diamond\phi \to \Box\Diamond\phi$	Euclidean

While $2^5 = 32$ subsets are possible, key collapse results (e.g., T implies D; T + 5 yields equivalence) reduce them to exactly **16 distinct logics**, all of which are formalized with full soundness and completeness.

```
          S5 = KTB45
         /    |    \
       S4    KTB   KB4
        \     |     /
         KT  KB   K4
          \  |  /
            K         ... + D and 5 extensions
```

Project Structure

```
ModalLogic/
├── Syntax/
│   ├── Formula.lean            — Formula type, complexity, substitution
│   ├── Derived.lean            — Derived operators (¬, ∨, ◇, ↔) and
notation
│   ├── ProofSystem.lean        — Hilbert-style proof system K
│   ├── BasicLemmas.lean        — Weakening, modus ponens, necessitation
│   └── SyntaxLemmas.lean       — Derived rule toolkit
├── Semantics/
│   ├── Semantics.lean          — Frames, models, forcing, validity
│   ├── Soundness.lean          — Soundness theorem (all 16 logics)
│   ├── Correspondence.lean     — Frame correspondence theory
│   ├── Definability.lean       — Definable frame properties
│   ├── Undefinability.lean     — Undefinability of irreflexivity
│   ├── Paths.lean              — Reachability and path theory
│   ├── FiniteModelProperty.lean — FMP via filtration
│   ├── Decidability.lean       — Decidability of K-validity
│   └── LocalConsequence.lean   — Local vs. global consequence
├── Metatheory/
│   ├── Maximal.lean            — Maximal consistent sets, Lindenbaum's
lemma
│   ├── Canonical.lean          — Canonical model and truth lemma
│   ├── CompletenessCube.lean    — Completeness for the classical cube
│   ├── CompletenessKDKB.lean   — Completeness for KD, KB
│   └── CompletenessD5.lean     — Completeness for remaining logics
└── Cube.lean                   — Unified modal cube with hierarchy
theorems
```

Additional files:

- **`Axioms.lean`** — Runs `#print axioms` on every major theorem to verify no `sorry` or unexpected axioms
- **`Examples.lean`** — Worked examples demonstrating the library

Quick Start

Prerequisites

- Lean 4 (v4.28.0 or compatible)

- Mathlib4 (fetched automatically by Lake)

Build

```
git clone <this-repo>
cd modal-logic-lean4
lake build
```

Usage

```
import ModalLogic

open Modal

-- Define propositional variables
def p := Form.var 0
def q := Form.var 1

-- Prove theorems in K
example : ∅ ⊢K (p ⊃ p) := ProofK.identity
example : ∅ ⊢K (□(p ⊃ q) ⊃ (□p ⊃ □q)) := ProofK.kdist

-- Use necessitation
example (h : ∅ ⊢K p) : ∅ ⊢K □p := ProofK.nec h
```

Proof Technique

Completeness is established via the **canonical model method**:

1. Assume a formula ϕ is not provable ($\nvdash \phi$)
2. Show $\{\neg\phi\}$ is consistent
3. Extend to a maximal consistent set via **Lindenbaum's lemma** (using Zorn's lemma)
4. Construct the **canonical model** where worlds are maximal consistent sets
5. Prove the **truth lemma**: $w \Vdash \psi$ iff $\psi \in w$
6. Obtain a countermodel, so ϕ is not valid ($\nvDash \phi$)

For logics beyond K, the canonical model's accessibility relation is shown to satisfy the required frame conditions (reflexivity, transitivity, etc.).

References

- Blackburn, de Rijke, and Venema. *Modal Logic*. Cambridge University Press, 2001.
- Chellas, Brian F. *Modal Logic: An Introduction*. Cambridge University Press, 1980.
- Hughes and Cresswell. *A New Introduction to Modal Logic*. Routledge, 1996.

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